



DEVELOPMENTX 1.0

Intelligent Edge-Monitoring for Early Fire Prediction, Gas Leak Detection, and Wildlife Safety

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System Overview Introduction

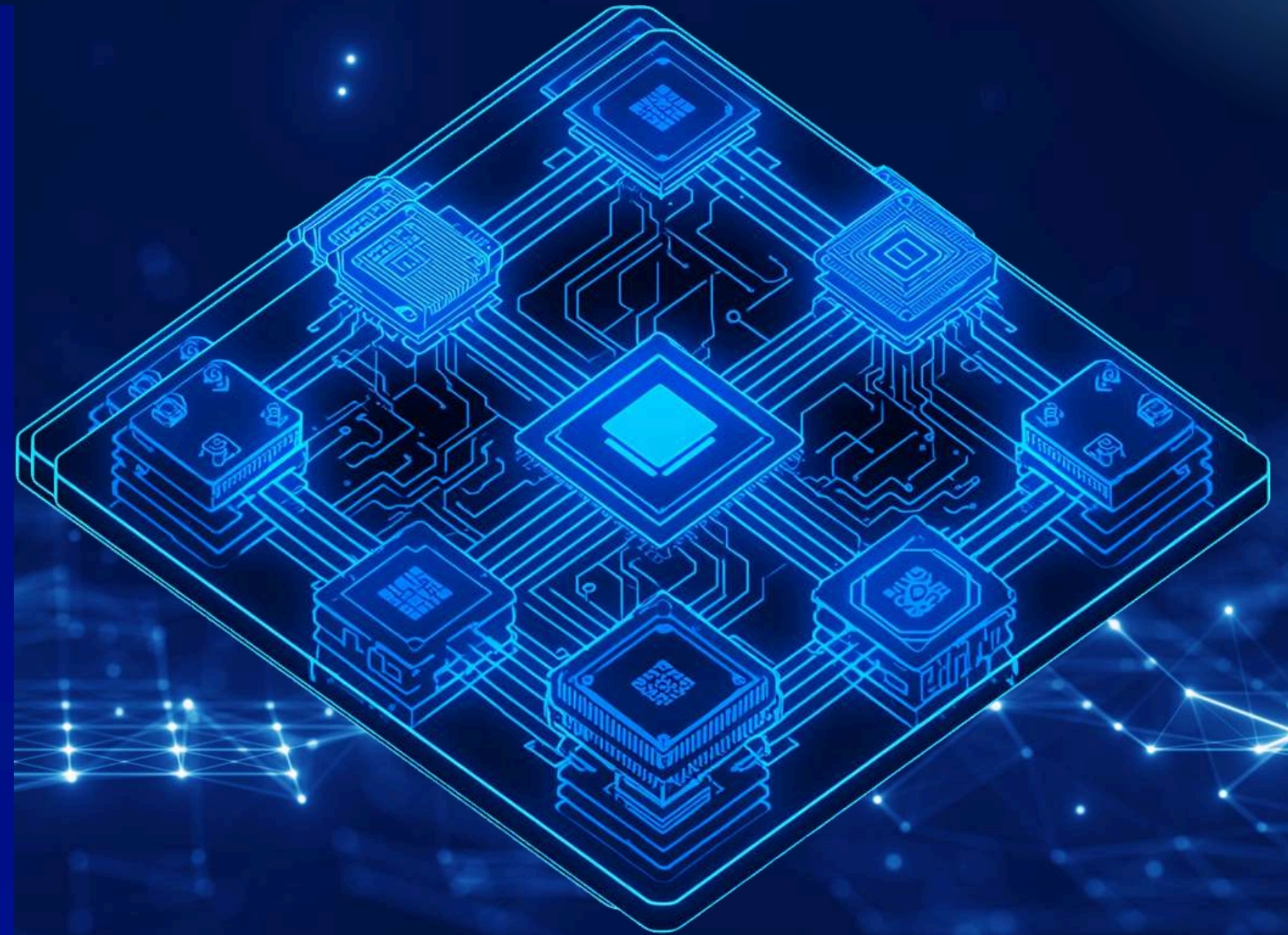
developmentX 1.0 is an autonomous, AI-driven IoT system engineered to proactively prevent industrial gas leaks, forest fires, and human-wildlife conflicts. Traditional monitoring relies on delayed human observation, often resulting in devastating consequences. To solve this latency, developmentX 1.0 acts as an intelligent perimeter shield that shifts emergency management from reaction to prevention.

At the hardware edge, an Arduino controller integrates with MQ gas detectors, flame sensors, and ultrasonic tripwires to monitor the physical environment. If a threat is detected, a SIM800L module instantly dispatches offline emergency SMS alerts.

However, the true command center of the system is the AI Live Monitoring Dashboard. Powered by an ESP32 wireless gateway, this centralized web interface provides remote, real-time control over the entire system. The dashboard visualizes raw sensor telemetry into live, readable gauges that track combustible gas levels (PPM) and heat indexes. Crucially, it features a Live AI Vision Console that streams continuous video from the ESP32-CAM. By utilizing object detection models, the dashboard actively draws real-time bounding boxes with confidence scores around encroaching wild animals or early-stage smoke signatures on the video feed.

The Problem – Critical Safety Challenges

Unnoticed gas leaks, rapidly spreading wildfires, and dangerous human-wildlife encounters cause devastating loss of life and environmental damage. Traditional monitoring relies on delayed human observation, often resulting in catastrophic consequences that could have been prevented with early detection systems.



Gas Leak Disaster – The Bhopal Tragedy

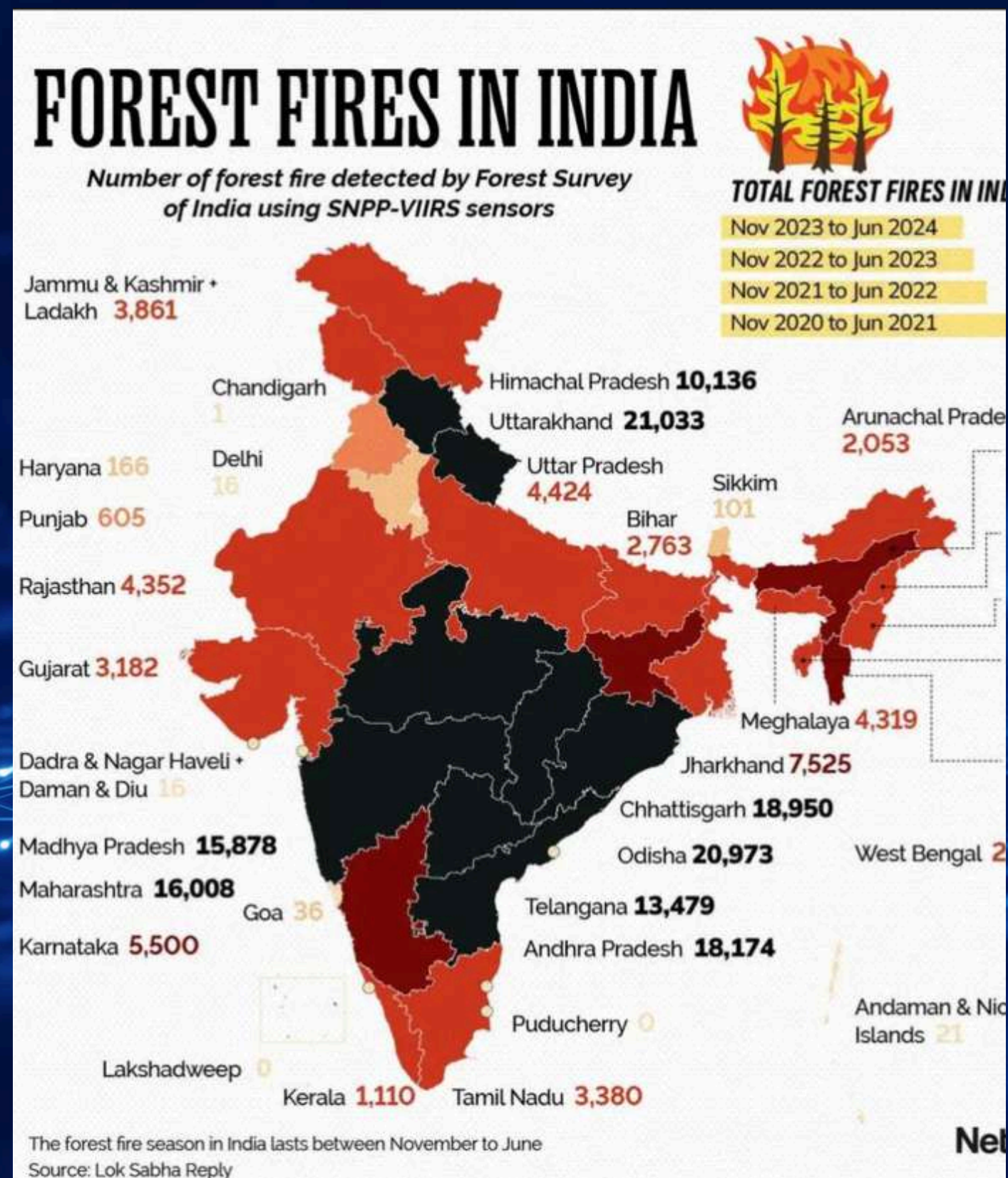
December 3, 1984: 40+ tons of MIC gas leaked from Union Carbide plant. 3,787 immediate deaths with estimates reaching 25,000+. Over 500,000 people exposed. Root causes: poor maintenance and lack of early warning systems. This catastrophe proves that early gas detection saves lives.

BHOPAL GAS TRAGEDY

DECEMBER 3, 1984



A tribute
to the innocent lives lost
in the world's worst industrial
disaster.



Forest Fire Impact – India & Global Statistics

India faces 30,000–35,000 forest fire incidents annually, with 2.67 million hectares lost in 2021 alone. Globally, 340–370 million hectares burn yearly, causing \$50+ billion in economic losses. 95% of fires are human-caused, yet early detection can reduce spread by 80%.

Human-Wildlife Conflict – Facts and Figures

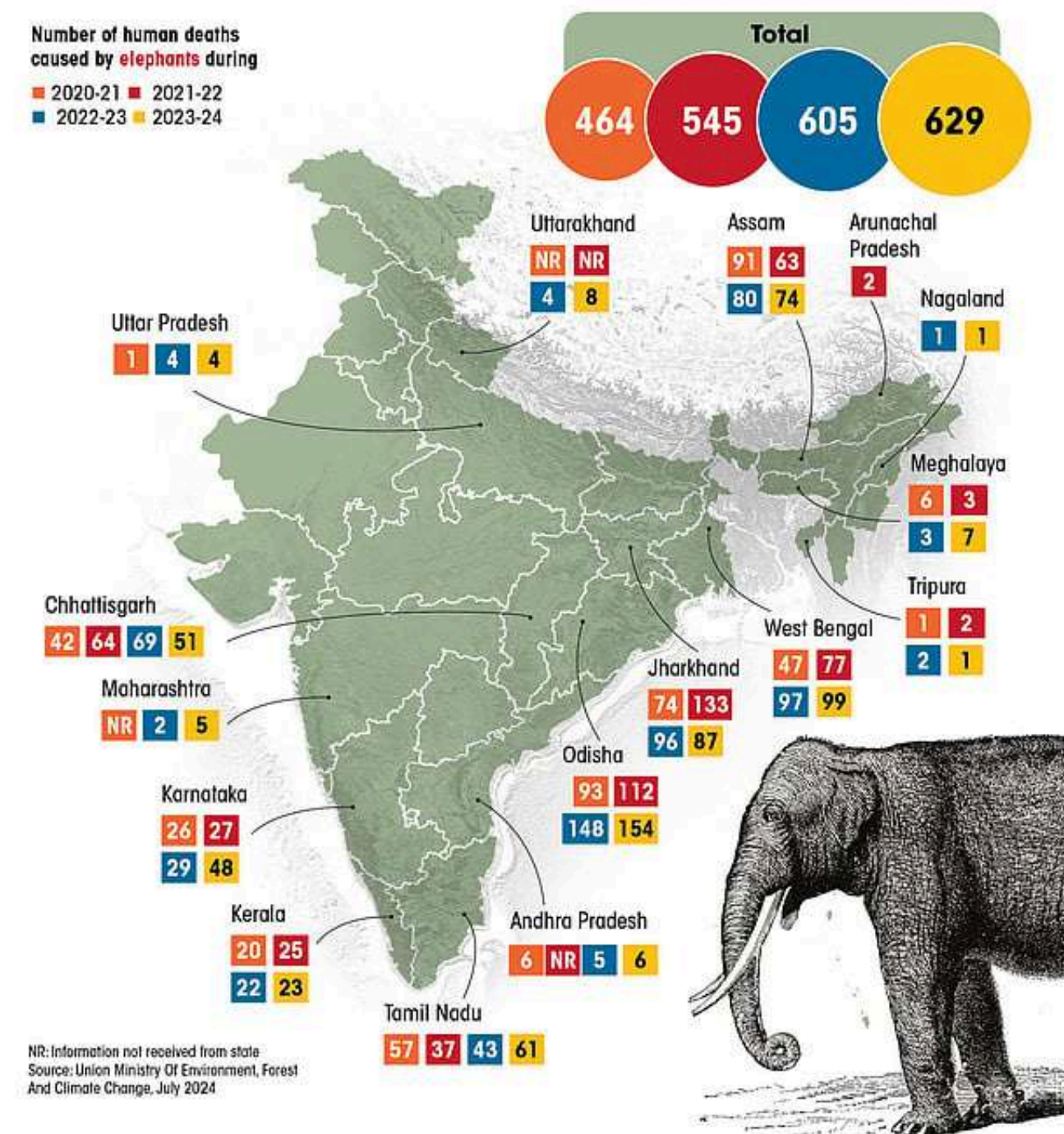
India faces over 50,000–60,000 human deaths annually from wildlife encounters. Elephant-human conflicts alone claim 400–500 lives yearly, while crop damage exceeds \$100 million. Early warning systems like developmentX 1.0 can reduce these conflicts by 60–70%.

Human-elephant conflicts

Human fatalities caused by elephant attacks increased in at least 10 states between 2020-21 (April to March) and 2023-24, with an overall rise of approximately 36 per cent. In 2023-24, three states—Odisha (154 deaths), West Bengal (99 deaths) and Jharkhand (87 deaths)—accounted for more than half of all deaths resulting from elephant attacks

Number of human deaths caused by elephants during

■ 2020-21 ■ 2021-22
■ 2022-23 ■ 2023-24



THE SMART SOLUTION

developmentX 1.0 is a comprehensive automated perimeter shield combining AI vision with environmental sensors. It detects gas leaks, early-stage fires, and tracks animal movement in real-time, instantly alerting authorities via a live AI-powered dashboard.





Hardware & Sensor Components

ESP32-CAM provides live video for AI detection. ESP32 serves as the Wi-Fi/Bluetooth gateway for data transmission. Arduino acts as master controller for sensor management. SIM800L GSM module sends emergency SMS alerts. MQ-2/MQ-135 sensors detect gas and smoke levels. Infrared flame sensor identifies fire signatures. Ultrasonic sensor tracks wildlife movement. Buzzer provides local audio alarms.



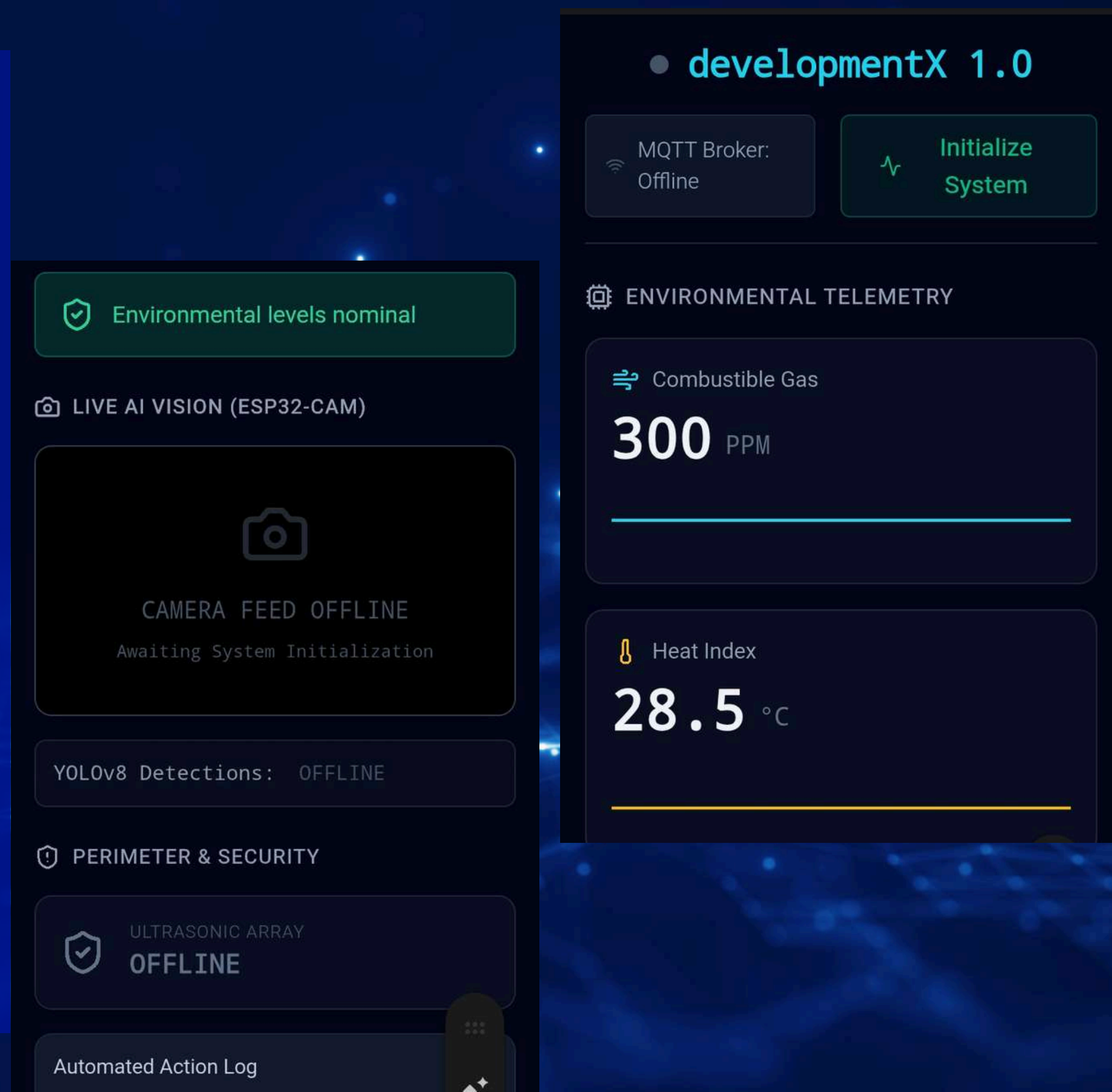
System Workflow & Alerts

Sensors collect environment data → Arduino manages inputs and triggers buzzer → ESP32 Gateway transmits data/video to AI dashboard → AI analyzes data for threats → SIM800L sends SMS alerts → Emergency teams respond rapidly.



AI Dashboard – Live Monitoring

The AI Dashboard features a Live Vision Console with real-time bounding boxes detecting smoke, fire, and animals. Environmental Telemetry Panel displays gas concentration and heat index. Perimeter Status Tracker monitors ultrasonic breach alerts with timestamps.



Benefits and Impact

Real-time, AI-enhanced detection reduces disaster risk significantly. The system protects human lives and wildlife ecosystems simultaneously through continuous monitoring. Automated alerts enable rapid, coordinated emergency responses. Scalable and adaptable for diverse environments and industries. Supports community safety and environmental sustainability.



Real-World Validation & Future Potential

Early detection systems demonstrate 80% reduction in fire spread when caught early, and 60–70% decrease in human-wildlife conflicts. Integration with regional disaster management and conservation programs enables coordinated emergency response. Future roadmap includes enhanced AI models, advanced sensor capabilities, and expanded geographic coverage.



Next Steps & Implementation

Successful deployment requires strategic planning: comprehensive site assessment and sensor network installation, hands-on training for local authorities and emergency responders, seamless integration with existing monitoring infrastructure, and establishing ongoing maintenance protocols with data analytics feedback loops.

Thank You

Git hub 
Google Drive 

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